CHEMISTRY PRACTICAL QUESTION MSCE

IDENTIFYING COVALENT AND IONIC SUBSTANCES USING MELTING POINTS

Materials: identical evaporating dishes (2), substances A and B (one is covalent and the other one is ionic but not necessarily in that order), source of heat, spatula.

Procedure: a. put a spatulaful of substance A into one evaporating dish and heat it for 1 minute.

Note whether it melts or not and record the results in the table of results.

b. put a spatulaful of substance B into the other evaporating basin and heat for 1 minute. Note whether it melts or not and record your results in the table of results

RESULTS

From the results, the substance which has melted / melted faster has low melting point and the

one which takes

time to melt / does not melt within the time limit given has high melting point.

Since covalent substances have low melting points and ionic substances have high melting points, it means that

the substance which has low melting point in the experiment is covalent and the one which has high melting

point is ionic.

2.IDENTIFYING COVALENT AND IONIC COMPOUNDS USING CONDUCTIVITY

Materials: conducting wires (3), an ammeter, beaker with holed rid, carbon rods (2), power

supply, solutions A and B (one is sodium chloride [common salt] solution and the

other is sucrose [common sugar] solution but not necessarily in that order).

Procedure: a. connect the circuit as shown in the figure below

b. read and record the ammeter reading in the appropriate space in the table of

results

c. replace solution A with solution B (remember to rinse the electrodes with distilled

water followed by solution B)

d. read and record the ammeter reading in the appropriate space in the table of

results

RESULTS

The solution which registers a reading is ionic and the one which does not register a reading is

covalent. This is so because ionic substances dissociate in water to give out ions which can conduct

electric current while covalent substances do not dissociate in water to give out ions which can conduct

electric current

If the solution conducts, it means that it is salt and if it does not conduct it means it is sugar. This is so

because salt is ionic and sugar is covalent.

RESULTS

The solution which registers a reading is ionic and the one which does not register a reading is

covalent. This is so because ionic substances dissociate in water to give out ions which can conduct

electric current while covalent substances do not dissociate in water to give out ions which can conduct

electric current

If the solution conducts, it means that it is salt and if it does not conduct it means it is sugar. This is so

because salt is ionic and sugar is covalent.

2.IDENTIFYING COVALENT AND IONIC COMPOUNDS USING CONDUCTIVITY

Materials: conducting wires (3), an ammeter, beaker with holed rid, carbon rods (2), power

supply, solutions A and B (one is sodium chloride [common salt] solution and the

other is sucrose [common sugar] solution but not necessarily in that order).

Procedure: a. connect the circuit as shown in the figure below

b. read and record the ammeter reading in the appropriate space in the table of results

c. replace solution A with solution B (remember to rinse the electrodes with distilled water followed by solution B)

d. read and record the ammeter reading in the appropriate space in the table of results

5.DISTINGUISHING AN ALKANE FROM AN ALKENE USING BROMINE TEST

Materials: liquids A and B in separate beakers (one is octane and the other one is octene but not necessarily in that order) and bromine in a dropper bottle.

Procedure: a. add a few drops of bromine in the beaker containing liquid A, observe what happens and record in the table of results.

b. add a few drops of bromine in the beaker containing liquid B, observe what

happens and record in the table of results.

RESULTS

The mixture that remains brown (the colour of bromine) contains octane and the one which turns

colourless contains octene. This is so because octene reacts with bromine to form dibromooctane

which is colourless while octane does not react with bromine so the bromine (brown in colour)

remains in solution.

7.SEPARATING A MIXTURE, W, OF SOLIDS X (SOLUBLE IN CARBON TETRACHLORIDE ONLY), Y

(SOLUBLE IN WATER ONLY) AND Z (INSOLUBLE IN BOTH CARBON TETRACHLORIDE AND WATER)

Materials: mixture W, distilled water, carbon tetrachloride, oven, filter funnel, filter paper, beaker,

evaporating dishes (3)

Procedure: a. dissolve W in distilled water

b. filter the mixture

c. wash the residue using distilled water

d. dry the filtrate by evaporation followed by drying in an oven

e. dry the residue by evaporation followed by drying in an oven

f. dissolve the well dried residue in carbon tetrachloride

g. filter the mixture

h. wash the residue using CCl_4

i. dry the filtrate by evaporation followed by drying in an oven

j. dry the residue by evaporation followed by drying in an oven

Pure solid Y is realized in step (d) since it is the one which dissolved in water.

RESULTS

Pure solid X is realized in step (i) since it is the one which dissolved in carbon tetrachloride.

Pure solid Z is realized in step (j) since it is the one which did not dissolve in any of the solvents

8.SEPARATING A MIXTURE OF SAND, NAPHTHALENE AND ALCOHOL

Materials: filter funnel, filter paper, beakers (2), the mixture of sand, naphthalene and alcohol.

Procedure: a. heat the mixture to 77oC to melt the naphthalene as shown below.

6.DISTINGUISHING ALKANE FROM ALKANOL USING SODIUM TEST

Materials: liquids A and B (one is pentane and the other one is ethanol), small pieces of

sodium metal (2), glass dish.

Procedure: a. pour a little of liquid A in a glass dish

b. drop a piece of sodium metal into the glass dish

c. observe what happens and record in the table of results by writing ‘reaction’ or ‘no

reaction’ where appropriate in the table.

d. repeat steps (a) to (c) for liquid B.

RESULTS

If the liquid reacts with sodium, it means the substance is ethanol, and if it does not react with sodium, the

substance is pentane.

This is so because alkanols react with metals

i.e. $C_2H_5OH + Na \rightarrow C_2H_5ONa + H_2$, while alkanes do not react with metals.

Note that the filtrate collected is a mixture of naphthalene and ethanol, hence sand has been separated from

the mixture.

c. let the filtrate cool for about10 to 15 minutes. Note that 2 layers are formed

since naphthalene is a solid and ethanol is a liquid at room temperature.

d. pour off the liquid into another beaker. This separates the 2 substances,

naphthalene and ethanol.